

Combinations — Problem Set

Note: While the focus of this problem set is Combinations, questions may involve Permutations as well.

Warm-up

1. Express nC_r in terms of factorials.
2. In how many ways can a committee of 3 be chosen from 10 people?
3. A bag contains 5 red marbles and 4 blue marbles. In how many ways can 2 marbles be chosen if there are no restrictions?
4. From a group of 8 students, one captain and one vice-captain are to be chosen. In how many ways can this occur?
5. Simplify $\frac{n!}{(n-1)!}$.
6. A menu offers 4 starters and 6 main courses. How many different combinations of one starter and one main course are possible?
7. How many ways can a set of 3 books be selected from a shelf of 9 different books?
8. A basketball team of 5 players is chosen from a squad of 12. How many different teams are possible?
9. In how many ways can 4 people be selected from a group of 10 if one specific person must always be included?
10. Simplify the expression $\frac{(n+1)!}{n!}$.
11. A committee of 3 is chosen from 8 people. Find the number of ways this can be done.
12. A bag contains 5 red and 4 blue marbles. In how many ways can 3 marbles be chosen if there are no restrictions?

Skill-building

1. A group consists of 5 boys and 7 girls. In how many ways can a team of 4 be chosen if it must contain exactly 2 boys and 2 girls?
2. Solve for n : $\frac{(n+1)!}{n!} = 8$.
3. In how many ways can 6 people be seated around a circular table?
4. A committee of 5 is chosen from 6 doctors and 4 nurses. How many committees contain at least 4 doctors?
5. A box contains 10 light bulbs, 3 of which are defective. If 2 bulbs are chosen at random, find the probability that both are defective.

6. How many different 3-digit numbers can be formed using the digits 1, 3, 5, 7, 9 if no digit is repeated?
7. A committee of 4 is to be formed from 10 people. If one specific person must be included, how many ways can the committee be formed?
8. A committee of 5 is to be formed from 6 doctors and 4 nurses. How many committees can be formed if:
 - a) There are no restrictions?
 - b) The committee must contain exactly 3 doctors?
 - c) The committee must contain at least 4 nurses?
9. A group of 10 people includes Alice and Bob. How many ways can a committee of 4 be formed if:
 - a) Alice must be on the committee?
 - b) Alice and Bob refuse to serve together?
10. A volleyball team of 6 is chosen from 10 players. One player is designated as the captain. How many ways can this be done?
11. In how many ways can 3 letters be chosen from the word **COMBINE** and then arranged in a row?
12. There are 7 points on a circle. How many distinct triangles can be formed using these points as vertices?

Easier Exam Questions

1. (Gosford 2020 Q10a)
 - (a) Find the number of ways that 4 students can be chosen from 6 juniors and 3 seniors
 - (i) without restriction. 1
 - (ii) so that there are unequal numbers of juniors and seniors. 2
2. (Blacktown Boys 2020 Q12a)

From a group of 7 cats and 8 dogs, 10 are chosen at random as a pet package to go to a nursing home.

 - i) In how many ways can this pet package be chosen? 1
 - ii) In how many ways can this pet package be chosen if there are exactly 6 cats and 4 dogs? 1
 - iii) What is the probability that this pet package will consist of at least 6 cats? 2
3. (Caringbah 2019 Q8)

A school music band consists of 8 dancers and a singer.

The dancers are selected from 17 students in Year 9.

The singer is selected from 5 students in Year 10.

In how many ways could the band be selected?

- (A) ${}^{17}C_8 + {}^5C_1$
- (B) ${}^{17}P_8 + {}^5P_1$
- (C) ${}^{17}C_8 \times {}^5C_1$
- (D) ${}^{17}P_8 \times {}^5P_1$

4. (Blacktown Boys 2023 Q12a)

From a group of 9 girls and 5 boys, they are chosen to form different patrol groups of 5 to monitor a busy pedestrian crossing.

- (i) If there are no restrictions, how many different groups of 5 are possible? 1
- (ii) What is the probability of choosing a group with 3 girls and 2 boys? 2
- (iii) If a group of 3 girls and 2 boys stand in a line, what is the probability that the boys stand together? 2

5. (Baulkham Hills 2025 Q3)

In how many ways can 7 people be chosen from a group of 16 people and then arranged in a circle?

- (A) $\frac{15!}{9!}$
- (B) $\frac{15!}{9!7!}$
- (C) $\frac{16!}{9!}$
- (D) $\frac{16!}{9!7}$

6. (Baulkham Hills 2023 Q11c)

A team of 4 players consists of 2 men and 2 women. If a total of 7 players are available, of which 3 are men and 4 are women;

- (i) How many different teams can be selected? 1
- (ii) If 2 of the players are husband and wife, and wish to play on the same team, how many different teams can be selected if both are chosen? 1

7. (Baulkham Hills 2020 Q11d)

A group of 12 students is to be divided into discussion groups.

- (i) In how many ways can the discussion groups be formed if there are 7 people in one group and 5 people in the other? 1
- (ii) In how many ways can the discussion groups be formed if there are 3 groups containing 4 people each? 2
- (iii) In how many ways can a president and vice president be chosen for the group of 12? 1

8. (Baulkham Hills 2022 Q11d)

A committee of five is to be formed from a group of 5 men and 9 women. In how many ways can this be done if:

- (i) there are no restrictions? 1
- (ii) the committee consists of a majority of women? 2
- (iii) two of the men refuse to serve on the same committee? 2

9. (Knox 2022 Q7)

Out of 12 contestants, eight are to be selected for the final round of a competition. Five of those eight will be placed 1st, 2nd, 3rd, 4th and 5th.

In how many ways can this process be carried out?

- (A) $\frac{12!}{4!5!}$
- (B) $\frac{12!}{4!3!}$
- (C) $\frac{12!}{8!}$
- (D) $\frac{12!}{8!5!}$

10. (Hurlstone 2024 Q11b & c)

(b) Three integers are chosen at random from the integers 1 to 20 (inclusive) In how many ways can these integers be chosen if their sum is even? **2**

(c) Solve the equation $16(n-1)! = 5n! + (n+1)!$ where n is any positive integer. **3**

Harder Exam Questions

1. (Glenwood 2020 Q7a)

There are five women and six men in a group. From this group, a committee of four is to be chosen that contains three women and one man.

- i) How many different ways can this committee be formed? **1**
- ii) A particular man and woman must be in the committee. What is the probability that this committee is formed? **1**

2. (Sydney Tech 2020 Q9c)

A team of six students is to be formed from a class of eight students. How many different teams can be formed if two particular students cannot be both selected for the team? **1**

3. (SCEGGS 2021 Q5b)

In a class of 23 students, there are 12 girls and 11 boys.

- (i) The class selects a captain and vice-captain. In how many ways is this possible if there are no restrictions? **1**
- (ii) A group of 4 students consisting of 2 girls and 2 boys is chosen at random from the 23 students to go on the student representative council.
 - (α) In how many ways could this be done? **1**
 - (β) What is the probability that a certain girl, Alana, and a certain boy, Brian, have been included in the student representative council group? **2**

4. (Knox 2021 Q12b)

From a group of 6 men and 7 women, a committee of 5 people is to be formed.

- (i) How many committees can be formed if there are more men than women? **1**
- (ii) What is the probability that if the committee contains three men and two women, that only one of Anne or Harold is definitely in the committee? **1**

5. (Presbyterian 2021 Q6c)
A seven-person committee is to be chosen at random from 5 primary school students and 7 high school students.
- (i) In how many ways can this committee be chosen? 1
 - (ii) What is the probability that the committee will consist of exactly 4 high school students? 1
6. (Gosford 2022 Q3c)
A committee of 3 is to be elected from a club of 8 members:
- i. How many different committees can be formed? 1
 - ii. If there are 4 women in the club, what is the probability that a randomly selected committee of 3 contains only women? 2
7. (Knox 2023 Q11b)
From a group of 6 men and 7 women, a committee of 5 people is to be formed.
- (i) How many committees can be formed if there are more men than women? 1
 - (ii) What is the probability that if the committee contains three men and two women, that only one of Anne or Harold is definitely in the committee? 1
8. (Glenwood 2019 Q19)
There are 8 girls and 7 boys in a class.
- (a) The class elects a captain and a vice captain. In how many ways is this possible? 1
 - (b) The class elects four representatives for the student council. How many different groups of representatives are possible? 1
 - (c) How many groups of representatives for the student council are possible if the class decides that they want two girls and two boys in the group? 1
9. (North Sydney Boys 2023 Section C Q2)
Four digit numbers are formed from the digits 1, 2, 3 and 4. 1
Each digit is used once only. The sum of all the numbers that can be formed is?
- A) 6666
 - B) 44440
 - C) 66660
 - D) 266640
10. (Glenwood 2024 Q8e)
Solve the following where n is any positive integer: 2
- $$16(n - 1)! = 5n! + (n + 1)!$$
11. (Hurlstone 2019 Q8 c & d)
c) Twenty students, ten male and ten female, are travelling from school to the pool for the annual swimming carnival. Twelve of them go in a minibus, six of them on bikes and two of them walk.

- (i) Write a mathematical expression that will give the number of ways the students can be divided for the trip. **1**
- (ii) Show all working to give the numerical answer for the number of ways the students can be distributed, if none of the boys walk. **2**
- d) A committee of 5 members is to be selected at random from 5 Australians and 3 New Zealanders. Determine the probability that there will be a majority of Australians on the committee. Show all working out. **3**

12. (Girraween 2024 Q8 a,c & e)

- (a) Write $\frac{1}{(n+1)!} - \frac{1}{(n-1)!}$ as a single fraction **2**
- (c) A five-person team is to be chosen at random from eight women and nine men.
- i. In how many ways can this team be chosen? **1**
- ii. What is the probability that the team will consist of five men? **1**
- (e) In how many ways can a committee of 5 be selected from 10 people, consisting of 5 men and 5 women, if one man, Ken, refuses to work in the same committee with Sue? **2**

13. (Baulkham Hills 2024 Q14c)

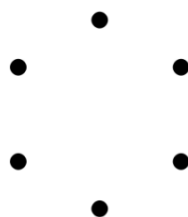
- (i) Show that ${}^{n+1}C_r - {}^nC_r = {}^nC_{r-1}$, for $r = 1, 2, 3, \dots, n$. **2**
- (ii) Hence find the value of ${}^3C_2 + {}^4C_2 + {}^5C_2 + \dots + {}^{100}C_2$. **2**

14. (Sydney Girls 2021 Q2d)

- 4 Maths books are chosen from 6 different Maths books and 3 English books are chosen from 5 different English books.
- (i) How many different ways can 7 books be selected? **1**
- (ii) Given that the 7 books have been selected, how many ways can they be arranged so that the Maths books stay together and the English books stay together? **1**

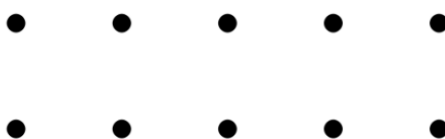
15. (Presbyterian 2023 Q14a)

- (i) Consider the dots below. **1**



Explain why 6C_3 distinct triangles can be formed using the dots as vertices.

- (ii) How many distinct triangles can be formed using the dots below as vertices? **2**



16. (SCEGGS 2023 Q10f)

It is given that

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

Do not prove this.

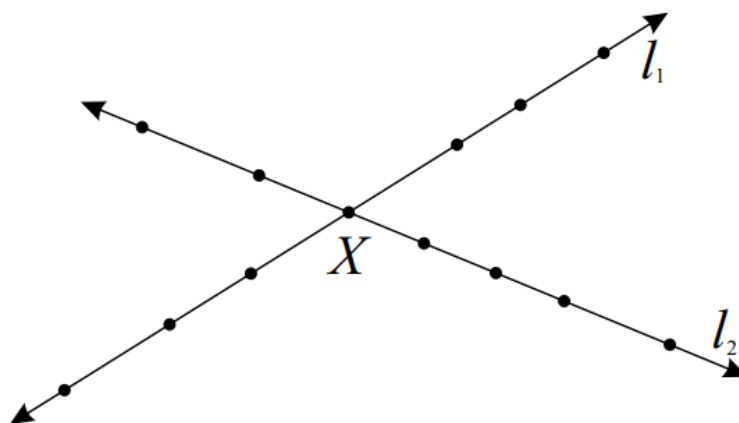
There are 23 students in a class. All 23 students are to be randomly divided up into two groups. There must be at least one student in a group.

How many ways is this possible?

2

17. (Sydney Grammar 2019 Q14b)

Note: This question is not for the faint of heart.



The diagram above shows two distinct intersecting lines l_1 and l_2 lying on a plane. Each line has n distinct points marked on it, with the point X lying on both lines. The case above shows a possible situation when $n = 7$, and there are a total of thirteen points marked.

Consider the general case where there are n distinct points marked on each line, for $n \geq 3$.

(i) In how many ways can three points be chosen from l_1 ?

1

(ii) By considering the total number of points on both lines and your answer to part (i), determine the number of triangles that can be formed using the marked points on either line as vertices.

2

(iii) Of the total number of possible triangles in part (ii):

(α) how many have X as a vertex?

1

(β) how many have two vertices that lie on l_1 ?

1

(iv) Use your answers to part (ii) and part (iii) to show that ${}^{2n-1}C_3 = 2 \times {}^nC_3 + (n-1)^3$.

2